

SAYAN DUTTA

sayandutta.developer@gmail.com | +91 8653128442 | Kolkata, West Bengal

[GitHub](#) | [LinkedIn](#)

SUMMARY

Computer Science undergraduate (BS-MS, IIT Patna) with hands-on software engineering experience building production-style backend systems, REST APIs, and applied ML pipelines. Independently designed and shipped a 30+ module Python backend with 1,900+ passing automated tests, and deployed a full-stack ML web application end-to-end. Comfortable across the stack: API design, data pipelines, model integration, and test-driven development.

TECHNICAL SKILLS

Languages: Python, C++, C, Java

Backend & APIs: FastAPI, REST API design, Pydantic, SQLite, MySQL, sqlite-vec, FAISS

ML / Applied AI: PyTorch, Hugging Face Transformers, Scikit-learn, Prompt Engineering, RAG, Google Gemini API

Engineering Practices: Automated testing, modular architecture design, dependency injection, ablation/experiment harnesses, Git

Tools & Platforms: Docker, Linux, CUDA, Google Cloud, Jupyter, Streamlit

PROJECTS

Blix – Modular AI Agent Backend Platform

2024-Present

Independent Project | Python, PyTorch, FastAPI, Pydantic, sqlite-vec | github.com/SAYANDUTTA8442/blix.ai

- Architected a 30+ module Python backend for an AI agent platform, exposed through a FastAPI service layer with 20 REST endpoints for memory, planning, and reasoning subsystems.
- Built and maintained an automated test suite of 1,962 passing tests, enabling confident refactors across a large, modular codebase.
- Designed a gradient-free adaptive tuning layer (contextual bandits) that adjusts system parameters from live usage signals in production, with automatic rollback on performance regression — no retraining or downtime required.
- Implemented a dependency-injected component framework allowing individual subsystems to be swapped, mocked, or stubbed independently for isolated testing and controlled experiments.
- Built a document ingestion pipeline (PDF, DOCX, OCR) to ground the system on external, unstructured data sources.
- Built a custom evaluation/monitoring harness tracking system health, retrieval accuracy, and task success rate — used to catch regressions before they reached production behavior.

ASES – AI-Driven Precision Agriculture Advisory Platform (Capstone)

Jan 2026 – June 2026

Python, Scikit-learn, Google Gemini API, Streamlit, SQLite | Live: sayanduttaiitpatnacda107.streamlit.app

- Shipped a full-stack web application end-to-end — from ML model to deployed product — with a Streamlit frontend and SQLite backend for persistent storage.
- Built and deployed KNN-based recommendation models over multi-dimensional soil and environmental data for crop and fertilizer suggestions.
- Integrated the Google Gemini API via structured prompting to generate natural-language advisory content on top of model predictions.
- Designed a modular pipeline separating prediction, recommendation, and advisory-generation layers for easy extension and maintenance.

EXPERIENCE

Research Intern — BITS Pilani (Remote)

Apr 2026 - May 2026

Empathetic Dialogue Generation System | PI: Dr. Srishti Gupta, Dept. of Computer Science

- Built and fine-tuned a T5-based NLP pipeline using FP16 mixed-precision training for a production-style empathetic response generation system.
- Automated an experimentation pipeline that ran 16 ablation studies to measure the impact of individual model components, cutting manual evaluation effort.
- Implemented data preparation tooling to construct preference-ranked training pairs used for model alignment.
- Collaborated with a 6-person research team to deliver results supporting a manuscript submitted to ACL ARR 2026.

EDUCATION

Indian Institute of Technology, Patna (IIT)

2025 – 2030 (Expected)

BS-MS, Computer Science and Data Analytics (Integrated 5-Year) | CPI: 8.64 / 10.0

Relevant coursework: Data Structures & Algorithms, Statistics for Data Science, Machine Learning, Numerical Methods, Data Analytics, Computer Science Fundamentals, Mathematics, Python Programming, Operating System, Database Management System, Computer Networks.

LEADERSHIP & ADDITIONAL INFORMATION

- Lead Google Student Ambassador, IIT Patna (2025–Present) — coordinated AI/ML workshops and technical outreach events for 100+ students.
- Co-author, ACL ARR 2026 submission on structured reasoning for dialogue generation (under review).
- Google Cloud Skills Boost: 70 completions including Introduction to Large Language Models, Generative AI, and MLOps for AI agents.